C4.2 Transfers of energy and matter: data-based questions

NAME	DATE	TOTAL
		_____ / 8

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 How fast are nutrients recycled?

8 MARKS



Image: Bernard Spragg. NZ from Christchurch, New Zealand, Public domain, Wikimedia Commons.

The mean residence time (MRT) is the average time a nutrient stays in dead organic matter, such as leaf litter, before decomposers release it back to the soil for plants. The table shows MRTs in four biomes.

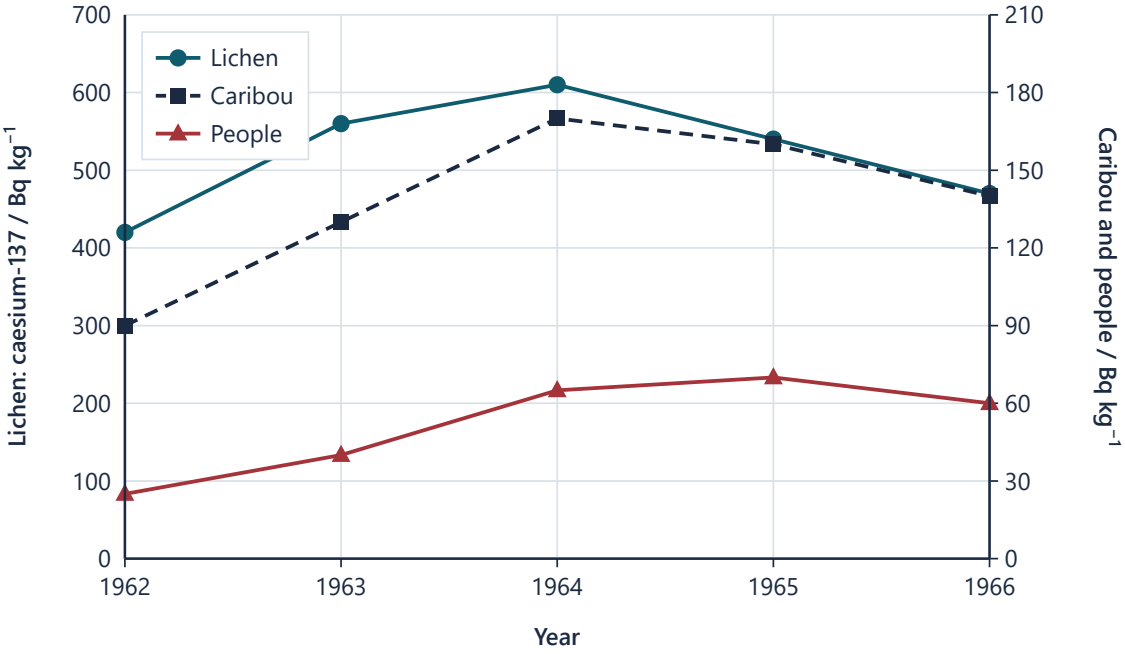
MRT / YEARS	BOREAL FOREST	TEMPERATE FOREST	CHAPARRAL (SCRUB)	TROPICAL RAINFOREST
NITROGEN	230	5.5	3.8	0.4
PHOSPHORUS	320	5.8	3.6	0.6
POTASSIUM	94	1.3	1.4	0.2

a IDENTIFY the biome in which nutrients are recycled fastest. [1]

b COMPARE the MRTs in the temperate forest and the chaparral. [2]

c SUGGEST two reasons for the very long MRTs in boreal forest. [2]

Radioactive caesium-137 from weapons tests settled on Arctic land in the early 1960s. It was measured in lichen, in caribou (which eat lichen) and in people who eat caribou.



d **DEDUCE** the trophic level of the lichen and of the people. [2]

e **SUGGEST** why caesium-137 in people peaked a year after it peaked in lichen. [1]